TI TIDA-060029 analog front end

Matched RG/RF: 100R / 5k / 100k

File: afe.kicad_sch

Codec DAC excitation

Coherent codec DAC excitation

File: source.kicad_sch

Codec acquisition (TAC5112)

Two channels / one frame clock

File: codec.kicad_sch

Range relays

Relay-selected ranges

File: range.kicad_sch

USB, RP2040 and boot / debug

USB control, firmware and host calibration

File: digital.kicad_sch

Power: 5V input, forced PWM, quiet analog rails

External 5V → quiet +/-12V

File: power.kicad_sch

Engineering prototype. One PCB. Isolated, discharged DUT only.

Temper

Sheet: /
File: temper-lcr.kicad_sch

Title: Temper LCR — TIDA-060029 based

Size: A3	Date: 2026-10-20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 7/7

U6A source buffer; U6B HI sense; U7A transimpedance; U7B LO sense.
Ranges: latching relays K1/K2/K3 (range sheet) close one matched RG/RF pair.
Four-terminal BNC jacks: HCUR = DUT_HI, LCUR = DUT_LO; followers sense HPOT / LPOT.
One local C104=100pF compensates all ranges; C78 100pF DUT_LO to GND (Rev E stability).
No BUF634A: maximum source current approximately 18mA. Isolated, discharged DUT only.

TI source and range network

U6A
OPA2810IDGKR
EXCITE
SRC_DRV
R104
47R
SOURCE_BUF
Rev E: R104 47R isolates the SOURCE_BUF relay bus from U6A.
R40
RG100
100R 0.1%
R41
RG5K
5k 0.1%
R42
RG100K
100k 0.1%
DUT_HI
J3
BNC HCUR
J7
R105
1k
HPOT
J8
BNC HPOT
R106
1k
LPOT
J9
BNC LPOT
DUT_LO
BNC LCUR
J3 HCUR, J7 HPOT, J8 LPOT, J9 LCUR: BNC, 22 mm pitch, shells to GND.
Connect an external, isolated, discharged DUT with Kelvin leads.
R105/R106 1k tie each POT to its CUR for two-terminal leads.

TI transimpedance topology; Rev C compensation
C104=100pF directly across U7.1 and U7.2 in all ranges

Rev D: R94 22R 0.1% between LCUR and the TIA summing node.
Rev E: LOW follower senses LPOT; host uses Zf = R94 + RF||C104.
R94
DUT_LO
22R 0.1%
TIA_SUM
R43
RF100
100R 0.1%
R44
RF5K
5k 0.1%
R45
RF100K
100k 0.1%
C104
100pF C0G 5%
U7A
GND
OPA2810IDGKR
TIA_OUT
C78
100pF C0G 5%
Rev E: C78 100pF C0G from LCUR (DUT_LO) to GND (TIA margin with relay parasitics).

U6B
OPA2810IDGKR
HPOT
HIGH_BUF
U7B
OPA2810IDGKR
LPOT
LOW_BUF

Independent HI / LO voltage followers

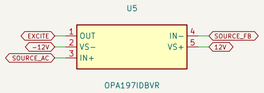
U6C
OPA2810IDGKR
12V
-12V
C42
100nF 50V
C44
1uF 50V
C50
10nF 50V
C43
100nF 50V
C45
1uF 50V
C51
10nF 50V

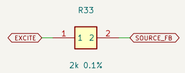
U7C
OPA2810IDGKR
C46
100nF 50V
C48
1uF 50V
C52
10nF 50V
C47
100nF 50V
C49
1uF 50V
C53
10nF 50V

Retain TI 100nF + 1uF bypass per rail; added 10nF at each supply pin.
PCB plane windows and short feedback routes are specified in reference-conformance.md.

Temper
Sheet: /TI TIDA-060029 analog front end/
File: afe.kicad_sch
Title: TI TIDA-060029 analog front end
Size: A2 | Date: 2026-10-20 | **Rev: E**
KiCad E.D.A. 10.0.4 | Id: 2/7

TAC5112 OUT1P/OUT1M differential line out at fs = 187.5kHz.
OPA810 difference stage: 2.2k in, 2.0k || 280pF (2 x 560pF in series) feedback and ground leg, gain 0.909.
DC coupled: the stage rejects the 1.65V DAC common mode.



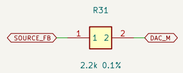




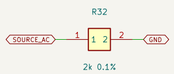




















Global net names connect functional sheets. See TI conformance matrix.

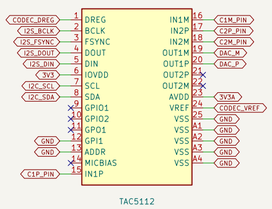
Temper

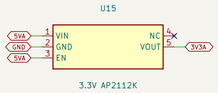
Sheet: /Codec DAC excitation/
File: source.kicad_sch

Title: Codec DAC excitation

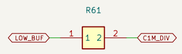
Size: A2	Date: 2026-10-20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 3/7

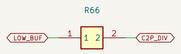
CH1 = HIGH_BUF − LOW_BUF (IN1P/IN1M); CH2 = LOW_BUF − TIA_OUT (IN2P/IN2M).
Each leg: 1k series, 1k + 560pF to GND (K = 0.5), 1uF, BAT54S clamp, 470R to the pin.
TDM controller: RP2040 drives BCLK 12MHz / FSYNC 187.5kHz. I2C address 0x50.
AVDD from U15 AP2112K−3.3 on 5VA; IOVDD from 3V3.

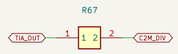




















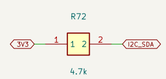


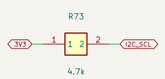


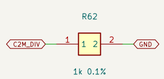








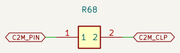


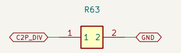
















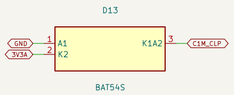






















Global net names connect functional sheets. See TI conformance matrix.

Temper

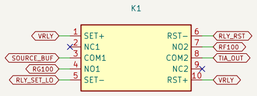
Sheet: /Codec acquisition (TAC5112)/
File: codec.kicad_sch

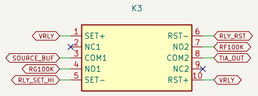
Title: Codec acquisition (TAC5112)

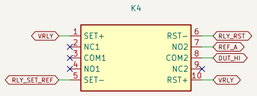
Size: A2	Date: 2026−10−20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 4/7

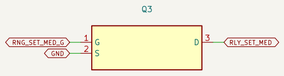
K1 100R, K2 5k, K3 100k, K4 reference: Panasonic TQ2SA-L2-5V dual-coil latching DPDT.
Pole 1: SOURCE_BUF -> RGx. Pole 2: TIA_OUT -> RFx. NC contacts unconnected.
RESET coils (K1-K4) in parallel on Q2; SET coils on Q3/Q4/Q5/Q6. 1N4148WS flyback per drain.
K4 reference: COM2 on HCUR, NO2 through R107 1k 0.1% to LCUR (REFCAPTURE, leads unplugged).
VRLY = 5V through R101 2.2R + C112 22uF. Firmware: source off, RESET 20ms, SET 20ms.



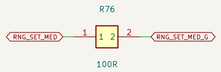


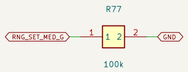




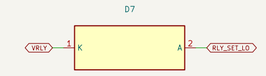


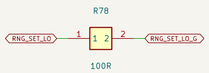


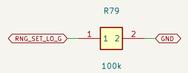


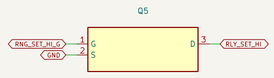


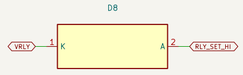


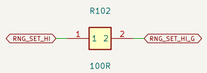


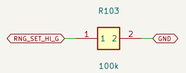


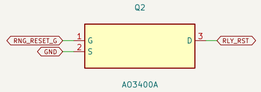




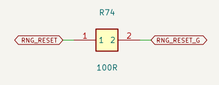


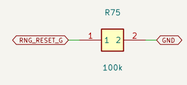


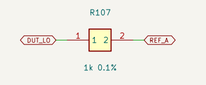


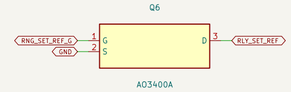


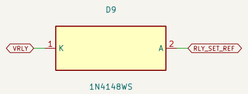


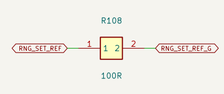


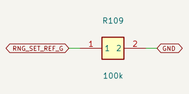


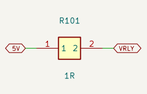














Global net names connect functional sheets. See TI conformance matrix.

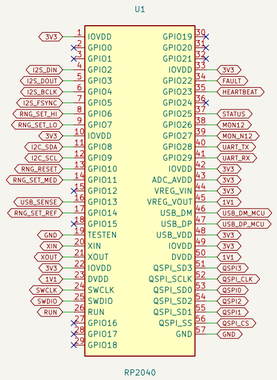
Temper

Sheet: /Range relays/
File: range.kicad_sch

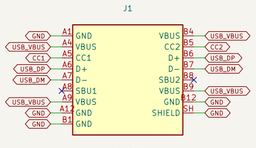
Title: Range relays

Size: A2	Date: 2026-10-20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 5/7

External 5V input powers the instrument. USB VBUS is sensed only; no backfeed.
Native USB CDC; 120MHz MCU; dual-lane ADC acquisition at up to 1MSPS.
Hold BOOTSEL while resetting for ROM USB firmware loading. SWD and UART provided.

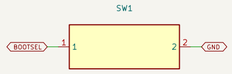


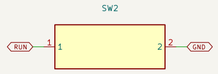


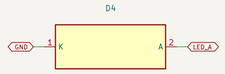








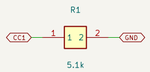


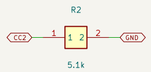


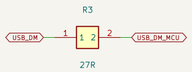


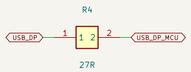


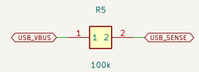


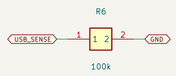


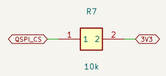


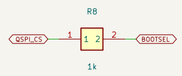


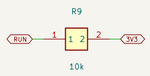


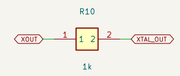


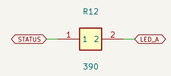


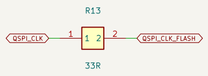




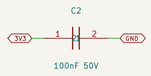




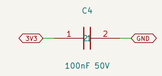




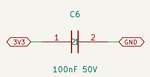




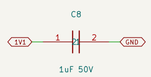










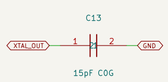




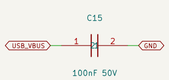


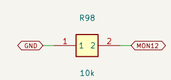


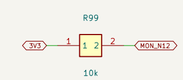






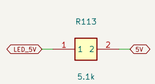


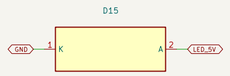


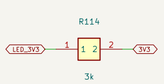


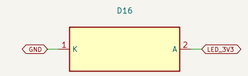


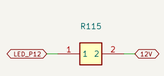




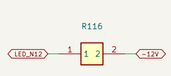




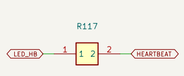




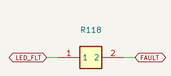














Global net names connect functional sheets. See TI conformance matrix.

Temper

Sheet: /USB, RP2040 and boot / debug/
File: digital.kicad_sch

Title: USB, RP2040 and boot / debug

Size: A2	Date: 2026-10-20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 6/7

Input: REGULATED 5V +/−5%, 1A, center positive. USB does not supply this board.
TPS65131 PSP/PSN grounded: forced PWM 1.25−1.5MHz. RAW_P ~14.31V: RAW_N ~−14.39V.
TPS7A49 ~+11.97V: TPS7A30 ~−12.00V. Fixed output values are nominal.
Do not substitute capacitor voltage/package without checking DC−bias derating.

U3
5V
GND
5V
1
2
3
VIN
GND
EN
NC
VOUT
4
5
3V3
3.3V AP2112K

U11
12V
REGP_FB
GND
LDOIN_P
1
2
3
4
5
OUT
FB
NC
GND
EN
NR/SS
DNC
IN
EP
6
7
8
9
REGP_NR
LDOIN_P
GND
TPS7A4901DGNR

U12
−12V
REGN_FB
GND
LDOIN_N
1
2
3
4
5
OUT
FB
NC
GND
EN
NR/SS
DNC
IN
EP
6
7
8
9
REGN_NR
LDOIN_N
GND
TPS7A3001DGNR

U13
SW_P
GND
GND
CONV_VIN
5V
5V
CONV_ENP
GND
CONV_ENN
GND
SW_N
1
2
3
4
5
6
8
9
10
11
12
13
INP
PGND
PGND
VIN
INN
BSW
ENP
PSP
ENN
PSN
NC
OUTN
OUTN
VNEG
FBN
VREF
CN
AGND
NC
CP
FBP
VPOS
INP
EP
14
15
16
17
18
19
20
21
22
23
24
25
SW_N
RAW_N
CONV_FBN
CONV_REF
CONV_CN
GND
CONV_CP
CONV_FBP
RAW_P
SW_P
GND
TPS65131RGER

U16
USB_VBUS
GND
USB_VBUS
1
2
3
IN
GND
EN
FAULT
ILIM
OUT
4
5
6
VBUS_ILIM
5V
TPS2553

D2
RAW_P
1
K
A
2
SW_P
SS14

D3
SW_N
1
K
A
2
RAW_N
SS14

L1
5V
1
1
2
2
SW_P
4.7uH 3.8A shielded

L2
SW_N
1
1
2
2
GND
4.7uH 3.8A shielded

FB1
5V
1
1
2
2
5VA
100R@100MHz 800mA

R95
RAW_P
1
1
2
2
LDOIN_P
1R

R96
RAW_N
1
1
2
2
LDOIN_N
1R

R80
5V
1
1
2
2
CONV_VIN
100R

R81
RAW_P
1
1
2
2
CONV_FBP
430k

R82
CONV_FBP
1
1
2
2
GND
40.2k

R83
RAW_N
1
1
2
2
CONV_FBN_MID
240k

R84
CONV_FBN
1
1
2
2
CONV_REF
150k

R85
12V
1
1
2
2
REGP_FB
82k

R86
REGP_FB
1
1
2
2
GND
9.1k

R87
−12V
1
1
2
2
REGN_FB
100k 0.1%

R88
REGN_DIV
1
1
2
2
REGN_DIV2
1k 0.1%

R89
REGN_FB
1
1
2
2
REGN_DIV
5k 0.1%

C16
5V
1
1
2
2
GND
1uF 50V

C17
3V3
1
1
2
2
GND
1uF 50V

C18
3V3
1
1
2
2
GND
10uF 25V X5R

C80
5V
1
1
2
2
GND
10uF 50V X7R

C81
CONV_VIN
1
1
2
2
GND
100nF 50V

C82
5V
1
1
2
2
GND
10uF 50V X7R

C83
5V
1
1
2
2
GND
100nF 50V

C84
CONV_REF
1
1
2
2
GND
220nF 25V

C85
CONV_CN
1
1
2
2
GND
4.7nF 50V

C86
CONV_CP
1
1
2
2
GND
10nF 50V

C87
RAW_P
1
1
2
2
CONV_FBP
15pF C0G

C88
RAW_N
1
1
2
2
CONV_FBN
4.7pF C0G

C89
RAW_P
1
1
2
2
GND
10uF 50V X7R

C90
RAW_P
1
1
2
2
GND
10uF 50V X7R

C91
RAW_N
1
1
2
2
GND
10uF 50V X7R

C92
RAW_N
1
1
2
2
GND
10uF 50V X7R

C93
REGP_NR
1
1
2
2
GND
10nF 50V

C94
LDOIN_P
1
1
2
2
GND
10uF 50V X7R

C95
12V
1
1
2
2
GND
10uF 50V X7R

C96
12V
1
1
2
2
REGP_FB
10nF 50V

C97
LDOIN_P
1
1
2
2
GND
100nF 50V

C98
REGN_NR
1
1
2
2
GND
10nF 50V

C99
LDOIN_N
1
1
2
2
GND
10uF 50V X7R

C100
−12V
1
1
2
2
GND
10uF 50V X7R

C101
−12V
1
1
2
2
REGN_FB
10nF 50V

C102
LDOIN_N
1
1
2
2
GND
100nF 50V

C103
5VA
1
1
2
2
GND
22uF 25V X5R

R97
MON12
1
1
2
2
12V
100k

R100
MON_N12
1
1
2
2
−12V
150k

R110
VBUS_ILIM
1
1
2
2
GND
24k 1%

C113
USB_VBUS
1
1
2
2
GND
4.7uF 25V X5R

PWR_FLAG
GND
PWR_FLAG
5VA
PWR_FLAG
RAW_N
PWR_FLAG
RAW_P
PWR_FLAG
LDOIN_P
PWR_FLAG
USB_VBUS

R112
REGN_DIV2
1
1
2
2
GND
5k 0.1%

R111
CONV_FBN
1
1
2
2
CONV_FBN_MID
1.5M

R119
CONV_ENP
CONV_VIN
1
1
2
2
3V3
100k

R120
CONV_ENN
1
1
2
2
3V3
100k

C116
GND
1
1
2
2
SEQ_EN
100nF 50V

C118
GND
1
1
2
2
5VA
100nF 50V

C117
5VA
1
1
2
2
GND
22uF 25V X5R

Global net names connect functional sheets. See TI conformance matrix.

Temper	
Sheet: /Power: 5V input, forced PWM, quiet analog rails/	
File: power.kicad_sch	
Title: Power: 5V input, forced PWM, quiet analog rails	
Size: A2	Date: 2026−10−20
KiCad E.D.A. 10.0.4	**Rev: E**
	Id: 7/7